

McCall's Search Model Beyond Pareto: When Does Easier Meeting Reduce Job Finding?

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1 Introduction

In the job-search model of McCall [1970], an increase in the meeting rate f has two opposing effects on the rate at which unemployed workers find jobs: offers arrive more often, but workers become pickier because hunting for a better offer is easier. Martellini and Menzio [2020] show that when the wage-offer distribution is Pareto, the two forces exactly offset, leaving the UE rate invariant to f . This knife-edge result rationalizes the absence of any secular trend in the job-finding rate despite large improvements in meeting technology.

This note asks a more general question: *for which offer distributions does the pickiness effect dominate*, so that

$$\frac{d \ln UE}{d \ln f} < 0?$$

The main result is a sharp characterization: the sign of the elasticity is governed entirely by the local elasticity of the *mean residual wage* $m(x) \equiv \mathbb{E}[w - x \mid w \geq x]$,

$$\text{sign} \frac{d \ln UE}{d \ln f} = \text{sign} \left(1 - \left. \frac{d \ln m(w)}{d \ln w} \right|_{w=w_R} \right).$$

The Pareto distribution is precisely the knife-edge case in which m is unit-elastic, which delivers the Martellini–Menzio invariance result as a corollary. Distributions with thinner tails (exponential, lognormal) have sublinear m and imply a UE rate *increasing* in f ; a UE rate *decreasing* in f requires tails heavier than any Pareto with finite mean—tail index exactly one with a slowly varying correction. We construct an explicit example, a Pareto distribution with a logarithmic correction, for which the UE rate is strictly decreasing in f everywhere, with a closed-form elasticity.

2 Model

2.1 Environment

Time is continuous. Workers are risk neutral and discount the future at rate r . An unemployed worker enjoys a flow payoff of zero ($b = 0$) and receives job offers at Poisson rate f .¹ Each offer is a wage w drawn from a distribution $G(w)$, i.i.d. over time, with finite mean. Offers must be accepted or rejected on the spot (no recall). An employed worker earns her wage w until the job is destroyed, which occurs at Poisson rate s , returning her to unemployment.

Throughout, $S(w) \equiv 1 - G(w)$ denotes the survival function of the offer distribution.

¹The zero flow payoff is for simplicity; all results below extend, with identical sign characterization, to a benefit proportional to the average accepted wage, $b = \bar{b} \mathbb{E}[w \mid w \geq w_R]$ with $\bar{b} \in [0, 1)$, as in Martellini and Menzio [2020].

2.2 Value Functions and the Reservation Wage

Let U denote the value of unemployment and $E(w)$ the value of employment at wage w . They satisfy the flow Bellman equations

$$rU = f \int \max\{E(w) - U, 0\} dG(w), \quad (1)$$

$$rE(w) = w + s(U - E(w)). \quad (2)$$

Equation (2) gives $E(w) = (w + sU)/(r + s)$, which is strictly increasing in w , so the optimal policy is a reservation-wage rule: accept if and only if $w \geq w_R$, where $E(w_R) = U$, i.e.,

$$\frac{w_R + sU}{r + s} = U.$$

Substituting into (1) and eliminating U yields the reservation-wage equation

$$w_R = \frac{f}{r + s} \int_{w_R}^{\infty} (w - w_R) dG(w). \quad (3)$$

The left-hand side is the flow benefit of accepting a marginal offer; the right-hand side is the option value of continued search.

2.3 The UE Rate

The rate at which workers transition from unemployment to employment is

$$UE = f S(w_R),$$

and its elasticity with respect to the meeting rate decomposes as

$$\frac{d \ln UE}{d \ln f} = \underbrace{1}_{\text{direct effect}} - \underbrace{\frac{G'(w_R) w_R}{S(w_R)} \frac{d \ln w_R}{d \ln f}}_{\text{pickiness effect}}. \quad (4)$$

The question is when the second term exceeds the first.

3 Main Result: A Mean-Residual-Wage Characterization

Define

$$H(x) \equiv \int_x^{\infty} (w - x) dG(w) = \int_x^{\infty} S(w) dw,$$

where the second equality follows from integration by parts (using finiteness of the mean), and the *mean residual wage*

$$m(x) \equiv \mathbb{E}[w - x \mid w \geq x] = \frac{H(x)}{S(x)}.$$

Two identities will be used repeatedly. Differentiating $m = H/S$ and using $H' = -S$,

$$m'(x) = h(x) m(x) - 1, \quad h(x) \equiv \frac{G'(x)}{S(x)}, \quad (5)$$

where h is the hazard rate; and consequently the elasticity of the survival function is

$$\varepsilon(x) \equiv \frac{G'(x) x}{S(x)} = \frac{x(1 + m'(x))}{m(x)}. \quad (6)$$

Proposition 1. Let $\phi \equiv f/(r + s)$. Then

$$\frac{d \ln UE}{d \ln f} = \frac{1}{1 + \phi S(w_R)} \left(1 - \frac{d \ln m(w)}{d \ln w} \Big|_{w=w_R} \right), \quad (7)$$

and in particular

$$\frac{d \ln UE}{d \ln f} \leq 0 \iff \frac{d \ln m(w)}{d \ln w} \Big|_{w=w_R} \geq 1.$$

Proof. The reservation-wage equation (3) reads $w_R = \phi H(w_R)$. Differentiating with respect to $\ln f$ and using $H' = -S$,

$$\frac{d \ln w_R}{d \ln f} = \frac{\phi H(w_R)}{w_R (1 + \phi S(w_R))} = \frac{1}{1 + \phi S(w_R)}.$$

Substituting into (4) and using (6) together with $\phi S(w_R) = \phi H(w_R)/m(w_R) = w_R/m(w_R)$,

$$\frac{d \ln UE}{d \ln f} = 1 - \frac{\varepsilon(w_R)}{1 + \phi S(w_R)} = \frac{1 + w_R/m(w_R) - w_R (1 + m'(w_R))/m(w_R)}{1 + \phi S(w_R)} = \frac{1 - m'(w_R) w_R/m(w_R)}{1 + \phi S(w_R)},$$

which is (7). $\square$

The economics of (7) is transparent. The reservation wage rises with f in proportion to the option value of search, which is governed by the mean residual wage $m(w_R)$: how much better are the offers still on the table? If m is *unit-elastic*, the pickiness effect exactly absorbs the direct effect. If m is *superlinear*—the further up the tail, the disproportionately better the remaining offers—the option value of waiting rises so strongly with f that workers reject offers faster than they arrive.

3.1 The Pareto Benchmark

The Pareto distribution is exactly the knife-edge case of Proposition 1.

Corollary 1 (Martellini and Menzio, 2020). If $G(w) = 1 - (w/\underline{w})^{-\alpha}$ with $\alpha > 1$, then $m(w) = w/(\alpha - 1)$ is *unit-elastic*, so

$$\frac{d \ln UE}{d \ln f} = 0 \quad \text{and} \quad \frac{d \ln \mathbb{E}[w \mid w \geq w_R]}{d \ln f} = \frac{1}{\alpha} > 0.$$

Indeed, solving (3) explicitly in the Pareto case gives

$$w_R = \left(\frac{\phi}{\alpha - 1} \right)^{1/\alpha} \underline{w}, \quad UE = (\alpha - 1)(r + s),$$

so the UE rate is invariant to f at every point, not merely locally: improvements in meeting technology show up entirely in wages, never in job finding. This is the balanced-growth property emphasized by Martellini and Menzio [2020].

Conversely, distributions with tails thinner than Pareto—exponential, lognormal, or any distribution with eventually decreasing mean residual wage elasticity below one—have sublinear m , so Proposition 1 implies $\frac{d \ln UE}{d \ln f} > 0$: the direct effect dominates. By (7), a *negative* response of job finding to meeting efficiency requires tails heavier than Pareto.

4 A Distribution with a Strictly Decreasing UE Rate

Consider, for $w \geq \underline{w} > 1$ and $\beta > 1$, the Pareto distribution with a logarithmic correction:

$$G(w) = 1 - \frac{\underline{w}}{w} \left(\frac{\ln w}{\ln \underline{w}} \right)^\beta. \quad (8)$$

This is a valid distribution: $S(w) = \underline{w}(\ln \underline{w})^\beta w^{-1}(\ln w)^{-\beta}$ is decreasing for $w > 1$ (since $w(\ln w)^\beta$ is increasing there). Its tail index is 1, yet the mean is finite:

$$H(x) = \underline{w}(\ln \underline{w})^\beta \int_x^\infty \frac{dw}{w(\ln w)^\beta} = \frac{\underline{w}(\ln \underline{w})^\beta}{\beta - 1} (\ln x)^{1-\beta}, \quad \mathbb{E}[w] = \underline{w} \left(1 + \frac{\ln \underline{w}}{\beta - 1} \right) < \infty.$$

The mean residual wage is

$$m(x) = \frac{H(x)}{S(x)} = \frac{x \ln x}{\beta - 1}, \quad \frac{d \ln m}{d \ln x} = 1 + \frac{1}{\ln x} > 1 \quad \text{for all } x > 1,$$

strictly superlinear everywhere.

Proposition 2. *Under the offer distribution (8), the elasticity of the UE rate is, in closed form,*

$$\frac{d \ln UE}{d \ln f} = -\frac{1}{\ln w_R + \beta - 1} < 0 \quad \text{at every } f.$$

Proof. The reservation-wage equation (3) implies $\phi S(w_R) = w_R/m(w_R) = (\beta - 1)/\ln w_R$, and (7) gives

$$\frac{d \ln UE}{d \ln f} = \frac{-1/\ln w_R}{1 + (\beta - 1)/\ln w_R} = -\frac{1}{\ln w_R + \beta - 1}. \quad \square$$

As $f \rightarrow \infty$, $w_R \rightarrow \infty$ and the elasticity vanishes from below: the UE rate falls with f , ever more slowly. Easier meeting makes workers *more* than proportionally pickier, so unemployment duration lengthens even as offers arrive faster—while accepted wages improve.

Remark 1 (Borderline tails are necessary for a global result). *For any regularly varying tail with index $\alpha > 1$, $m(x) \sim x/(\alpha - 1)$ as $x \rightarrow \infty$, so $\frac{d \ln m}{d \ln x} \rightarrow 1$ and the UE elasticity converges to zero. A globally negative elasticity therefore forces a tail index of exactly 1 with a slowly varying correction, as in (8)—the heaviest tail consistent with a finite mean. If one only requires $\frac{d \ln UE}{d \ln f} < 0$ over a range of f , milder departures suffice: any distribution with $\frac{d \ln m}{d \ln w} > 1$ on an interval (e.g., suitable Pareto mixtures) delivers a negative elasticity whenever w_R lies in that interval.*

References

- Paolo Martellini and Guido Menzio. Declining search frictions, unemployment, and growth. *Journal of Political Economy*, 128(12):4387–4437, 2020.
- John J. McCall. Economics of information and job search. *Quarterly Journal of Economics*, 84(1): 113–126, 1970.